Pre-Checkride Oral Review

1. **Describe engines:** 2 continental IO-550 C, fuel injected engines, air cooled (IO: “I” = fuel injection, “O” = horizontally opposed
2. **HP:** 300, each engine
3. **Max rated RPM:** 2700
4. **Max stater time:** 30 seconds
5. **Fuel:**
    a. Type – 100LL (blue) or 100 (green)
    b. Total – 172 gal
    c. Total useable – 166 gal
    d. Main useable- 83 usable per side
6. **Oil Capacity:** 12 qts full (9 minimum, 10-11 normal)
7. **Max gross weight:** 5500
8. **Empty weight:** See POH (Complete W&B form based on your weight - DPE 170lbs, & 110 gal fuel)
9. **Airspeeds:**

a.	Vso	65	Dirty stall speed (flaps/gear extended)
b.	Vs1	78	Clean stall speed (flaps and gear retracted)
c.	VMC	84	Minimum control speed w/one engine failed (red line)
d.	Vx	92	Best angle of climb/minimal intentional 1 engine operation (yellow line)
e.	Vxse	95	Beast angle of clime – single engine
f.	Vy	105	Best rate of climb – both engines
g.	Vyse	101	Best rate of climb – single engine (blue line)
h.	Vfe	122	Max flap extended – 69-122 (white arc)
i.	Vle	152	Max gear extended
j.	Vr	85	Rotation speed
k.	Best glide		115
l.	Emer descent		152 (practice at 135k)
m.	Vsse	88	Minimum intentional one engine operation (yellow line)

10. **Single engine service ceiling:** Will not climb > 50 fpm
11. **Multi engine service ceiling:** Will not climb > 100 fpm
12. **Accelerate stop distance:** Distance required to accelerate to *Vr, bring throttles to idle, and come to full stop* (see performance chart in POH)
13. **Accelerated go distance:** Distance required to accelerate to *Vr, have engine fail, feather prop, and continue takeoff to 50 agl* (see performance chart in POH)

14. **Propeller system:** 3 blade constant speed prop. Springs & counterweights move blades to high pitch (less RPM at runup, less drag in flight), engine oil pressure moves blades to low pitch (higher RPM at runup – more drag in flight).

    Propeller controls on left side of engine control quadrant. In flight, aft levers in flight (high pitch) results in feathered condition (less drag). On takeoff, low pitch (levers full forward) create more thrust.

    Why do props not feather when oil pressure is lost due to engine shut down? Anti-feathering lock pins prevent the prop from feathering due to loss of oil pressure if engine shuts down.

15. **Electrical system:** 28v system (2-12v batteries wired in series to produce 24v to the 2 electrical busses. 2 belt driven 50 amp alternators (one per engine), controlled by dual voltage regulators. Only one regulator is functioning at any given time, the other acts as backup regulator.

    Electrical system provides power to starters for engine starting. Once started, magnetos provide spark to engines. Batteries/alternators provide power for starter, lights, radios, avionics, heater blower, cowl flaps, landing gear, etc.

16. **Vacuum system:** Suction for vacuum operated gyroscopic instruments is provided by 2 engine driven suction pumps (working together as a single system). If one fails, the other will continue to provide suction to instruments. Suction gauge is on instrument panel and has two pop-out red buttons that serve as source identifiers, one for each side of system. If button is popped out, suction pump has failed.

17. **Stall warning system:** Warning vane is located on leading edge of left wing. Vane is electrically operated and activates a horn in the cockpit when approaching stall. Stall warning is effective in all flight attitudes, weights, and airspeeds, but can be affected by icing.

18. **Fuel system:** Fuel system controls (between seats, at floor) provides an on/off/cross-feed system. 166 usable located in main tanks. Fuel gauges are on instrument panel. Mechanical fuel pump is attached to each engine and provides fuel to injectors. A 2-speed electrical fuel pump (hi/low/off) is on panel. High for starting (4 sec when cold) or in event of mechanical pump failure. Low is for eliminating fuel pressure fluctuations and restart after engine failure.

    Cross-feed procedure: level flight only, inoperative engine to off position, aux fuel pump on operative engine on LOW, good engine to cross-feed, then fuel pump off or low as required. A lockout system prevents both selectors from being in cross-feed or off position at same time.

19. **Landing gear system:** Landing gear is driven by electric motor and controlled by 2-position switch located on instrument panel. Gear switch in equipped with a green/red light to indicate gear status (red=up, green=down). There is also a mirror on the left engine cowling that provide a visual of the nose gear when extended and an indicator at floor level under center of the panel.

    A handle for manually operating the gear located behind the co-pilot seat. NEVER OPERATE THE GEAR ELECTRICALLY WITH THE MANUAL HAND CRANK ENGAGED, AS IT COULD SPIN AND CAUSE DAMAGE OR INJURY.

    3 instances of gear warning horn: 12 inches MP or less, gear selector in up position on ground (safety or squat switch prevents this), full flaps selected with gear not extended

Backup gear extension:

I. LDG OR MOTOR Circuit Breaker – PULL
2. Landing Gear Handle – DOWN
3. Remove cover at rear of co-pilot seat (helps to move seat forward). Engage hand crank and turn counterclockwise as far as possible (50 turns) disengage hand crank (helps to slide co-pilot seat forward)
4. If electrical system is operative, check landing gear position lights and warning horn (check LDG OR RELAY circuit breaker engaged)

20. **Brake system:** Wheel brakes are hydraulically operated and are located on the main struts. A hydraulic reservoir is located in the nose compartment. Fluid can be checked with dip stick located on the reservoir cap. Brake systems are independent and can be operated by pushing top of pilot's pedals (no brakes on co-pilot side).

21. **Flap system:** Wing flaps are electrically controlled and actuated. A three-position switch, up/approach/down lowers and raises the flaps. There is a two-light (red=up, green=down) above the flap switch.

22. **Pitot-static system:** The system is composed of 2 pitot tube, one the left nose, and one on the right nose and two static ports, one located on each side of the empennage. Pitot heat is provided via a switch on the instrument panel. The pitot/static system feeds the instrumentation.

23. **Emergency static port:** Where is it located? And how does it work? See placard for operation, located at pilot's position on lower left side (red placard)

24. **Anti-ice system:** pitot heat,.

25. **Heater:** How does it work? (janitrol combustion heater)

26. **Do all multi engine aircraft have a critical engine?** No. Aircraft with counter-rotating props do not have a critical engine. Loss of either engine has same aerodynamic affect.

27. **Which engine is the critical engine?** The engine, that when inoperative, has the greater adverse effect on control and performance

28. **Define critical engine factors:** P-factor (yaw), accelerated slipstream (roll & pitch), spiraling slipstream (yaw), torque (roll)

29. **Define VMC:** VMC is the minimum speed at which directional control can be maintained with the critical engine inoperative (Red-Line, 84kts published). In layman's terms, it is the speed below which control authority required to counteract the forces generated by the loss of the critical engine is no longer available. The manufacturer determines VMC based on FAA mandated conditions (be able to answer questions about VMC factors)

30. **Loss of engine-actions:** Bank, ball, blue line - mixture full, props full, throttles full, check flaps/gear up, (dead foot – dead engine), identify, verify, feather, mixture to lean, see emergency checklist for restart procedure.

31. **Manifold pressure:** It is the vacuum created by the engine on the intake stroke. It is a measure of the power that the engine is producing, via the amount of force the vacuum is creating.

32. **What affect is there on manifold pressure in a climb?** Manifold pressure decreases approximately 1 inch with every 1000 feet during a climb. Since the engine depends on the

density of air, it loses power with altitude, so the manifold pressure you can achieve at full throttle also decreases with altitude.

33. **If you are on approach and maintaining blue line, but not maintaining glide slope, what do you need to do?** Add power as required to fly level while maintaining blue line or above, recapture glide slope, then descend on glide slope and adjust power as needed to maintain glide slope/blue line or above
34. **Navigation system:** Garmin GTN™ 650Xi
35. **Autopilot:** stec
36. **Communications:** Com 1 = GTN™ 650Xi